map-charting-student-math-misunderstandings

Private 7th (Public 10th) Place Solution

Rank	7
Team	S T I T Y
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Topic ID	612134
Version	Original

Source: https://www.kaggle.com/competitions/map-charting-student-math-misunderstandings/discussion/612134
Retrieved with Kaggle CLI on 2026-10-21.

First, we would like to express our deepest gratitude to the organizers and Kaggle. This competition has been an incredible learning experience for our team.

Team Member: @yukitake93 @isakatsuyoshi @takai380 @sinchir0 @takaito

Overview

- **Direct Logit Training:** Assigned tokens to Misconception labels and trained directly on extracted logits using CrossEntropy loss
- **vLLM Integration:** While classification heads were tested with similar scores, we chose the logit-based approach for optimal vLLM inference performance
- **Ensemble Strategy:** Combined multiple pipelines with different models and prompting strategies

Accelerating Tips

1.Synthetic Data Generation (mainly by ISAKA)

To improve model accuracy, we generated synthetic data using GPT-4.1 and GPT-5-mini.

Synthetic Data Generation Method:

- Extracted data from the original train dataset meeting specific criteria: groups of 5+ data points with identical "Correct Label (True/False)", "Question ID", and "Misconception"
- Randomly sampled 3 entries from each group as reference to generate new "student explanations"
- Using 3 different entries created more diverse and higher-quality synthetic data (avoiding overfitting to single examples)

Generated Versions:

- `ver2-46`: Generated by GPT-4.1
- `ver47-56`: Generated by GPT-5-mini
- Dataset available here

Evaluation Results:
Used Chris Deotte's notebook "Gemma2-9B-it - CV 0.945" as baseline.

k=5 Results

Dataset	CV Score	LB Score
No synthetic dataset	0.940189	0.944
ver2-13	0.943963	0.946
ver14-25	0.943781	0.947
ver26-37	0.944208	0.949
ver26-37 (retrained)	0.943890	0.948
ver38-46	0.943645	0.947
ver47-56 (Created with GPT5-mini)	0.942850	0.946

k=6 Results

Dataset	CV Score	LB Score
No synthetic dataset	-	-
ver2-13	0.943990	0.947
ver14-25	0.944604	0.949
ver26-37	0.944294	0.947
ver38-46	0.943394	0.948
ver47-56 (Created with GPT5-mini)	0.944408	0.947

Improvements with synthetic data:

- **CV Score:** +0.003 to +0.004
- **LB Score:** +0.002 to +0.005

Important Notes:
These improvements were observed specifically with Gemma2-9B-it + OOF predictions. Other models showed mixed results, suggesting model-specific adaptations are crucial for effective synthetic data utilization.

2.Misconception Candidate Filtering

- Leveraged predetermined incorrect answer candidates for each question
- Added only relevant misconception candidates to prompts → significantly reduced prompt length

3. Noise Mitigation and Label Quality Enhancement

- **Label Smoothing:** Added α=0.1 to loss function for select models to handle noisy labels
- **Knowledge Distillation:** Trained Qwen2.5-32B on competition data and used outputs as soft labels (instead of using generated labels for synthesized data directly)

4. Pre-identifying Correct Answers

- As shared in this discussion, train and test data share the same 15 questions
- Focused solely on predicting Misconceptions, added True/False labels in post-processing

5.Alternative Answer Options in Prompts

- Added other MC answer options (excluding the student's selected answer) from the same question to provide additional context

What Not Worked

Dynamic Ensemble Weight Adjustment (mainly by takaito)

- Attempted to dynamically adjust ensemble model weights per question
- Limited to fold0 CV due to computational constraints
- Insufficient data likely caused poor performance
- Multiple fold CV might have improved results

Small Model Addition (mainly by takaito)

- Prepared lightweight models (minutes for prediction)
- No improvement when added to ensemble

Pipeline Details

Pipeline 1 (yukichi)

Data Strategy:

- Competition data + ISAKA's synthetic data(using v2-13)
- Two prompt types:
  - **Type 1:** Relevant misconception candidates + explanations reverse-engineered using Claude Sonnet (generated detailed explanations for each misconception based on actual student response patterns)
  - **Type 2:** Complete misconception catalog

Model Architecture:

Model	Prompts	Public Best	Private
Gemma-2-9B-IT	2	0.950	0.946
Qwen2.5-32B (AWQ)	2	0.950	0.947

Ensemble Result: Public LB 0.952, Private 0.948 🎯

Distillation Strategy:

Data Type	Strategy
Competition	Original labels (50%) + Qwen2.5-32B OOF (50%)
External	Qwen2.5-32B logits only

Regularization:

- **Gemma2-9B:** Label smoothing (α=0.1) + Temperature 1.5
- **Qwen-32B:** No smoothing + Temperature 1.0

Pipeline 2 (takai)

Data Strategy:

- Competition data + ISAKA-san's synthetic data
- Pre-generated misconception candidates via Claude Sonnet

Model:

- deepseek-ai/DeepSeek-R1-Distill-Qwen-32B (Public Best :0.950, Private: 0.945)
- Generally run inference on the dataset with models trained per fold and use the outputs as soft labels.
- Train with CE_loss + KL_div_loss on external data (using only items whose OOF top-1 prediction matches the label) plus competition data (Public: 0.95, Private: 0.945).
- Train with CE_loss + KL_div_loss on competition data (Public Best: 0.948, Private: 0.943).

Pipeline 3 (sinchir0)

Basically, I trained using the code from my public notebook.

Key factors that led to the improvement

- Rather than using the added special tokens, the existing tokens are used as the correct answers.(CV+0.005, LB+0.005)
- Instead of using the checkpoint with the lowest eval_loss, use a slightly later checkpoint. (CV+0.008,LB+0.007)

Data Strategy:

- Competition data only
- Potential misconception options only
- GPT-5 generated explanations for candidate misconceptions

Model:

- Qwen/Qwen3-8B (Public Best: 0.948, Private: 0.945)
- Full Train + Full Fine-Tuning
- lr: 2e-5 (1e-4 caused convergence issues)

- 3epoch

Key Takeaways

- Synthetic data effectiveness is highly model-dependent
- Prompt engineering and candidate filtering crucial for efficiency
- Combining distillation with label smoothing effectively handles noisy data
- Pre-identifying correct answers eliminates one source of error

Thank you for reading our solution!

Supplemental Q&A

Bilzard · 2025-10-17T03:34:37.820000

Congratulations for gold medals.

@isakatsuyoshi

It seems generation data have much effect especially on new data (i.e. LB). What number of examples generated in total?
And do you mind sharing example generation prompt?

ISAKA Tsuyoshi · 2025-10-17T03:47:01.013000

Thank you!
A total of 200,000 records were generated, and about 50,000 records from version 2-13 are used in the final solution.

I'll share the prompt later.

UPDATED:
As a design choice, I randomly sampled three reference data points for each generation.
This helped introduce diversity into the outputs.

```
prompt =

 j, (_, row)  (sampled.iterrows(), ):
    prompt +=

prompt +=
```

Bilzard · 2025-10-17T06:02:17.563000

> A total of 200,000 records were generated, and about 50,000 records from version 2-13 are used in the final solution.

That's quite an amount. What kind of filtering policy is used to select 50K examples?

ISAKA Tsuyoshi · 2025-10-17T09:09:51.193000

> Yes. I treated both the training and LB data as i.i.d., and since I assumed the misinterpretation patterns were finite, I generated a large amount of data to comprehensively cover all possible types.
>
> I used only one prompt, so all samples were drawn from the same probability distribution. I did not apply any special filtering. I believe the variation in results stems mainly from mismatches between the generated student explanations and the labels, rather than from data quality itself.
> This issue, I believe, was mitigated by the soft-label approach proposed by my teammate.